|INCH-POUND|

MIL-L-80277
29 September 1989

MILITARY SPECIFICATION

LATHES, PRECISION, TOOLROOM, VARIABLE SPEED,
11-INCH THROUGH 13-INCH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers 11-inch through 13-inch precision lathes having infinitely variable spindle speeds for producing parts to precision tolerances from a variety of materials with equipment and accessories specified herein. The requirements set forth in this specification and the completed Ordering Data (section 6.2) establishes the Government's minimum requirements for each procurement.

1.2 Classification. The lathes covered by this specification shall be of the following sizes. The size to be furnished shall be as specified (see 3.5 and 6.2).

Size	Swing over bed	Swing over cross slide
1	11 inch	6 inch
2	12 inch	7 inch
3	13 inch	8 inch

|Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, 2163 Airways Boulevard, Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.|

AMSC N/A FSC 3416

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interferences.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of Federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

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(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 390.03a - Gear Classification, Materials and Measuring Methods for Bevel, Hypoid, Fine Pitch Wormgearing and Racks Only as Unassembled Gears.

AGMA 2000-A88 - Gear Classification and Inspection Handbook.

AGMA 2001-B88 - Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth.

(Application for copies should be addressed to the American Gear Manufacturers' Association, Standards Department, 1500 King Street, Suite 201, Alexandria, VA 22314-2717.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B5.8 - Chucks and Chuck Jaws. (DOD adopted)

ANSI B5.9 - Spindle Noses for Toolroom Lathes, Engine Lathes, Turret Lathes, and Automatic Lathes. (DOD adopted)

ANSI B5.10 - Machine tapers. (DOD adopted)

ANSI B5.16 - Accuracy of Engine and Toolroom Lathes. (DOD adopted)

ANSI B11.6 - Safety Requirements for the Construction, Care, and Use of lathes. (DOD adopted)

ANSI/ISO 54 - Cylindrical Gears For General Engineering and for Heavy Engineering, Modules and Diametral Pitches of. (DOD adopted)

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

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NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA MG 1 - Motors and Generators. (DOD adopted)

(Any additional NEMA Standards.)

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete lathe shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Design. The machine shall be new (not a prototype) and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The lathe shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair. Neither a gap bed design, nor the use of a spacer between the headstock and the bed, shall be acceptable in order to achieve the capacity required in this specification.

3.2.1 Measuring and indicating device calibrations. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1), metric (see 6.5.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the lathe shall be graduated in the same system.

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3.2.1.1 Dials. Rotating dials or handwheels shall be used for longitudinal movement and shall be graduated in increments not larger than 0.005 inch. Rotating dials or handwheels shall be used for cross motion and shall be graduated in increments not greater than 0.001 inch. Such graduations shall accurately indicate machine component movement or material removal. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position. All feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.1.2 Digital readout system. Unless otherwise specified (see 6.2), a digital readout system shall be installed on the X and Z axes of travel. The system shall consist of a position feedback device, an electronic display for each axis, and all related components, mounting hardware, and electrical equipment necessary to ensure complete operation of the system. The system shall provide a readily visible digital display over the full working range of each axis on which installed. The X axis and Z axis resolution shall be 0.0001 inch. The system accuracy shall be 0.00050 inch verified upon installation and calibration of each system on the machine. The repeatability shall be 0.00010 inch, measured over the full travel on each axis. The X axis shall show twice the amount of cross slide movement to reflect the reduction of the workpiece diameter. Special requirements for the system components shall be as specified (see 6.2). The display shall indicate axis location in both the inch-pound and metric units. An input power source fluctuation of $\pm$ 10 percent shall not adversely affect the system operation.

3.2.1.2.1 Position feedback device. A position feedback device including all related components, hardware, and electrical equipment shall be provided for each axis having a readout system. The feedback device shall be of the independent type with a resolution equal to the resolution of the digital readout system. Each feedback device shall be installed with a cover or guard to prevent physical damage to any portion of the feedback device. The connecting cable from the position feedback device shall be electronically shielded to protect against electronic interference and shall be armored to protect against contamination and damage. The device shall provide axis travel feedback over the entire range of axis travel for each specified axis. Installation of the position feedback device shall not limit machine travel on any axis nor eliminate manual machine stops. Provisions shall be made to prevent damage to the feedback device in the event of overtravel. The feedback unit and related components shall be protected from contamination. Position feedback devices shall be easily and quickly interchangeable between like machines procured under the same contract.

3.2.1.2.2 Electronic display console. The electronic display console for the digital readout system shall contain a universal digital display with individual illuminated digits of not less than 0.43 inch in height. Digits shall be high efficiency displays for high light ambient conditions with wide

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viewing angle and shall be readable at a distance of not less than 16 feet. The console shall display at least six digits per axis simultaneously. The display console shall also provide keypad entry, memory, absolute/incremental counting, data preset, power failure indicator, inch/metric conversion, and absolute zero recall. The electronic display console shall be installed so as to be readily accessible and easily read from the operator's normal work station. The console shall be mounted to withstand shock and vibration generated by the machine. Each connecting cable shall be armored and shall be securely fastened to the structure of the machine. Location of installed cables shall prevent any interference with normal machine operation. Electronic display consoles shall be easily and quickly interchangeable between like machines procured under the same contract.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt, or remanufactured end products are allowed under this specification.

3.2.3 Energy efficiency. The lathe and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

3.2.4 Controls. All mechanical, hydraulic, and pneumatic operating controls shall be located convenient to the operator's work station(s).

3.2.5 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the lathe whether or not specifically required herein, shall conform to all requirements of OSHA 29 CFR 1910 and ANSI B11.6. If a conflict arises between 29 CFR 1910 and ANSI B11.6, 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2). Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial machine supplied to the commercial market shall be provided for the point of operation and all other parts of the lathe that present safety hazards. If point-of-operation guarding is influenced by workpiece size and configuration, the procuring activity shall provide part design data to the supplier of the machine (see 6.2). The supplier of the machine shall provide the special guarding which shall meet OSHA 29 CFR 1910.

3.2.6 Mercury restriction. The lathe shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Environmental protection. During the manufacture, operation, service, transportation, or storage of the machine and its components, materials

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hazardous to the ecological system as prohibited by applicable Federal, state, or local statutes in effect on the date of the contract shall not be used or emitted.

3.2.9 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a cleanable or replaceable filter. Means for determining oil level in each reservoir shall be by an oil level sight gauge, a low oil level warning light, or similar device and shall be in clear view of the operator's normal work station. An automatic lubrication system shall be provided for all ways and axis screws. Each lubricant reservoir shall have at least a 24-hour capacity and means for determining fluid level. All oil holes, grease fittings, and filler caps shall be readily accessible.

3.2.10 Interchangeability. All parts subject to replacement shall be manufactured to definite dimensions and tolerances. Replacement parts (except for hand-fitted parts such as ways and boxed bearings) shall be interchangeable without requiring modification. Replacement parts are those which are subject to wear or failure such as gears, bearings, shafts, feed screws, drive belts, pumps, and seals. Electrical replacement parts shall include, but not be limited to, motors, relays, fuses, switches, magnetics, pushbuttons, light bulbs, braking mechanisms, and solid-state devices. Tooling and accessories normally supplied as standard items shall also be considered replacement parts. Foreign-sourced replacement parts shall be interchangeable with domestically supplied components or be readily available from a domestic source.

3.3 Construction. The lathe shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion or failure.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

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3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2), the paint color and application shall be in accordance with the manufacturer's commercial practice. All surfaces to be painted shall be cleaned of all foreign matter prior to paint application.

3.3.6 Threads. All threaded parts used on the lathe and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Gears. All gears shall conform to or exceed all provisions of AGMA 390.03a, AGMA 2000-A88, and AGMA 2001-B88 for the inch-pound system of measurement or ANSI/ISO 54 for the metric system of measurement. The gears shall be manufactured of a suitable ferrous material. The gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears in the drive train shall be hardened and ground steel. Working surface hardness shall be not less than Rockwell C-48.

3.3.8 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The lathe shall consist essentially of a bed and base, headstock, headstock spindle, spindle drive, spindle speed change mechanism, carriage, apron, cross slide, compound rest, and tailstock. In addition, all equipment necessary for the machine to meet the performance requirements stated herein shall be provided.

3.4.1 Bed and base. The bed shall be a stress relieved ferrous casting designed to retain accurate way alignment. All bed way surfaces which support and guide the carriage shall have a hardness of not less than Rockwell C-50. Provision shall be made for using a taper attachment. The base shall be a heavy steel weldment or a casting providing rigid support for the lathe. The motor drive and variable speed mechanism may be mounted inside the base or on the rear side of the bed with suitable guard enclosures. The top of the base shall have a chip and coolant pan designed to drain coolant to a sump. The base shall be provided with means for securing and leveling the machine on a foundation.

3.4.3 Headstock. The lathe shall have a headstock that supports and aligns the spindle axis parallel with the bed ways. The headstock shall contain a clutch and a quick acting brake mechanism to start and stop the spindle rotation and a power takeoff to the gearbox for feeding and threading. Spindle start and stop shall be by a lever on the headstock of the lathe. The headstock shall

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support the spindle and gear shafts to prevent deflection that would affect accuracy under full operating and cutting loads. All load carrying surfaces of gears, clutch jaws, surfaces of shafts on which gears and clutches slide, or internal surfaces of sliding parts shall have a surface hardness of not less than Rockwell C-48. When power to the spindle drive mechanism is transmitted through V-belts, means shall be provided to maintain proper belt tension. Headstocks which contain oil shall be oiltight and shall have automatic lubrication to all internal parts that require lubrication. When specified (see 6.2), headstocks which contain oil shall be provided with a temperature sensor to monitor the temperature of the lubricant and shut off the machine at a preset temperature below the flashpoint of the oil vapor.

3.4.4 Headstock spindle. The lathe shall have a headstock spindle to provide accurate mounting and rotation of a chuck, faceplate, or collet. The spindle shall be supported by roller and ball type bearings of a size, type, and precision class suitable for the application. Unless otherwise specified, (see 6.2),the spindle nose shall be a type D-1 and shall conform to the requirements of table I and to ANSI B5.9. The spindle internal taper shall be as specified in table I and shall conform to the requirements of ANSI B5.10. The spindle nose shall be accurately machined to provide locating surfaces for lathe centers, faceplates, and chucks. The inner and outer surfaces of the spindle nose on which work holding devices are mounted shall be hardened to no less than Rockwell C-50.

3.4.5 Carriage. The carriage, cross slide, and compound slide shall be accurately fitted to the mating slide ways. The carriage cross slide and compound slide shall have adjustable antifriction rollers, full length adjustable gibs, or a similar device to compensate for wear. The carriage and cross slide shall be operated manually and by power without binding or chatter. The cross slide feed screw shall be ground, have a thread hardness of not less than Rockwell C-40, and have antifriction bearings. Way wipers shall be provided on slide units. The compound tool slide shall have a T-slot for mounting tool holders, shall swivel 360 degrees, and have a scale graduated around the base in 1 degree increments for angular settings. The carriage apron shall be of double-wall construction and shall support shafts within at both ends. A manually operated pressure lubrication system shall supply oil to the operating components of the carriage and to the bed ways. Lever type operating controls such as for threading and feed engagement shall fully engage and disengage with smooth positive action.

3.4.6 Spindle speed change mechanism. The spindle speed change mechanism shall be the variable speed type designed so that speed change cannot occur unintentionally. Speeds shall be selected by positioning a dial to permit spindle speed adjustments to compensate for varying machining conditions. It shall be possible to adjust the spindle speed while the spindle is rotating in either direction. The actual spindle speed in Revolutions Per Minute (RPM) shall be shown by a speed indicating device located convenient to the operator.

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3.4.7 Power feed. Power feeds and thread leads shall be provided through a totally enclosed quick change gearbox with feed rod, ground precision leadscrew, and a geared feed mechanism within the carriage apron. Longitudinal and cross feeds shall be in both directions of travel regardless of the direction of spindle rotation. The leadscrew and end gearing shall be reserved for thread cutting only. Automatic lubrication shall be provided for the quick change gearbox and for end gearing, when provided. The end gearing shall have provision for use of transposing gears to obtain metric threads, if a combination English and metric gearbox is not required. Safety clutches or other suitable devices shall protect the feed mechanism from damage by accidental overload. An interlock shall prevent simultaneous engagement of the thread and feed controls.

3.4.8 Variable Power feed. When specified (see 6.2), infinitely variable longitudinal and cross feeds shall be provided by a totally enclosed ball bearing type electric motor. Feed selections shall be made by a suitable control device showing chosen feeds in inches per minute, (ipm) or SI units, as required, and shall be adjustable while cutting. Feeds shall be in both directions of travel of the carriage and cross slide. The power feed shall be inoperative when the spindle is stopped, and when using the threading mechanism. Thread leads shall be provided by a totally enclosed quick change gearbox and a precision leadscrew which shall be reserved for thread cutting only. The end gearing shall have provision for use of transposing gears to obtain metric threads, if a combination English and metric gearbox is not required. Safety clutches or other suitable devices shall protect the feed mechanism from damage by accidental overload. An interlock shall prevent simultaneous engagement of the thread and feed controls.

3.4.9 Combination English and metric gearbox. Unless otherwise specified (see 6.2), the lathe shall be equipped with a selective-type combination English and metric gearbox to provide a full range of US and metric feeds and threads as specified in Table I, without the use of transposing gears. The thread selections shall include the U.S. Standard fine and coarse series and pipe threads. The compound slide feed screw shall be equipped with a dial graduated to read measurements in both the English and the metric systems. Threading shall be done using a leadscrew from the gearbox and a split nut mechanism within the carriage apron. The lathe shall be able to cut either a right-hand or left-hand threads regardless of the direction of spindle rotation. The gearbox shall be automatically lubricated. A feed and thread chart shall be mounted near the gearbox as a ready reference for feed and thread selections. All feed and thread selections shall be shown without need for mathematical conversion.

3.4.10 Tailstock. A tailstock shall be accurately fitted to the bed ways and aligned with the headstock spindle. Unless otherwise specified (see 6.2), the tailstock shall be marked to indicate a zero position as aligned with the headstock spindle centerline and to indicate a set-over of not less than one-half inch each side of zero. The spindle taper shall be in accordance with ANSI B5.10 for self-holding tapers and shall eject tools when fully

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retracted. The spindle shall have a hardness of not less than Rockwell C-38 and shall be graduated in increments not greater than 1/16 inch for measuring spindle extension. The spindle shall be easily extended and retracted by a handwheel to both limits of travel without jamming. The tailstock shall be easily moved along the bed ways and have a quick acting clamp for locking the tailstock to the bed and a device for locking the spindle in any extended position without deflection which would cause loss of the accuracy required in Table II. The spindle mechanism and tailstock guide ways shall be provided with means of lubrication.

3.4.11 Coolant system. The coolant system shall be a complete unit for delivering a flow of coolant to the tool. The coolant system shall include a sump or reservoir, a power driven pump, and all necessary piping. The reservoir shall have a positive means to prevent chips from entering the pump. The system shall provide means for draining and cleaning. All materials used in the coolant system shall be compatible for use with all soluble and standard cutting oils. An adequate supply of coolant shall be continuously available to the cutting tool at all locations within the work envelope. An adjustable, flexible outlet tube shall provide final dispensing of coolant. A manually operated flow regulating valve shall be located near the tube outlet to allow adjustment of the stream of coolant between off and full flow, as required by the operator. A coolant off-on switch shall be located convenient to the operator.

3.4.12 Electrical requirements. The machine shall be wired to conform to NFPA 79. Unless otherwise specified (see 6.2), each machine shall be designed to operate on 230/460-volts, 3-phase, 60 Hertz (Hz). Unless otherwise specified (see 6.2), the lathe shall be initially wired for operation on 460 volts. Each machine shall have a fused safety disconnect switch or circuit breaker. An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in NFPA 79 for the disconnect fuse rating. The wiring in each machine shall have individual point-to-point color coded wires in conformance with NFPA 79. Diagrams/schematics shall clearly identify the individual, point-to-point color coded wire and wire termination point location in the electrical system.

3.4.13 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2), alternating-current (ac) motors shall be designed to operate on 60-Hz. All motors shall conform to the requirements of NEMA MG-1 and the motor enclosure shall meet the NEMA MG-1 requirements for a drip proof enclosure. Each motor shall have an identification plate to identify manufacturer, model identification, serial number, voltage, amperage, horsepower, phase and frequency. Motor horsepower shall be the manufacturer's standard or standard optional horsepower as indicated on the manufacturer's machine design and construction drawing(s). It shall be unacceptable to substitute a motor larger than the manufacturer's standard to meet the requirements of table I.

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3.4.14 Worklight. Unless otherwise specified (see 6.2), a worklight shall be provided. The worklight shall be a complete unit with all necessary equipment for mounting on the carriage, shall have a protective shield, an off-on switch and shall be adjustable to provide the maximum lighting of the work area. The lamp shall be shielded against dirt and chips and shall operate on no more than 130 volts. If a built-in lighting system is supplied the worklight described shall not be required.

3.4.15 Hour meter. When specified (see 6.2), the lathe shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the lathe.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the lathe shall meet the requirements of table I for the machine specified. Machine capacity shall be not less than the stated requirements of the size ordered. When a range is given, the required performance shall be from the stated minimum or less to the stated maximum or greater.

3.5.1 Physical size limitations. When specified (see 6.2), the physical size of the machine shall not exceed the height, width, and length restrictions specified by the procuring activity to ensure the machine will fit in its future operating location.

3.6 Alignment tolerances. Each lathe shall meet the alignment and performance accuracies set forth in table II using the tests described in ANSI B 5.16, "Accuracies of Engine and Toolroom Lathes."

3.7 Performance. The lathe shall be capable of meeting all the performance requirements specified herein when tested in accordance with 4.7.1 and 4.7.4.

3.7.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2).

3.8 Standard equipment. All standard equipment required to properly operate the lathe (such as special tools) shall be furnished. The equipment specified in 3.8.1 through 3.8.8 shall also be furnished as standard.

3.8.1 Lathe centers and center sleeve. One headstock center and one tailstock center shall be furnished. One headstock center sleeve shall be provided when required to adapt the taper in the spindle nose to the taper of the center supplied for the tailstock. The centers shall be hardened to Rockwell C-62 to C-66. The centers and the sleeve shall be ground to within 0.0002 inch total indicator reading (tir) for concentricity over their entire lengths.

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3.8.2 Chuck and chuck jaws. The size and type of the chuck and the number of chuck jaws shall be as specified (see 6.2). The chuck shall be a wrench operated, manual chuck, constructed of steel and shall mount on the spindle nose without the need for an adapter. The top jaws shall be the reversible, stepped type for internal and external gripping and shall be hardened. A self-ejecting chuck key shall be provided for each chuck. In all other aspects, the chuck shall conform to the requirements of ANSI B5.8 for Class I, Medium Duty, Wrench Operated Chucks.

3.8.3 Taper attachment. The taper turning attachment shall be for mounting in any working position on the rear of the lathe bed. Unless otherwise specified (see 6.2) the taper attachment shall be graduated in inches of taper per foot and degrees. The attachment shall not interfere with normal lathe operation when not in use.

3.8.4 Tool post. The tool post shall be either a round screw type with a rocker base, a square indexing turret type for holding at least four tool holders, or a quick change tool post capable of holding a tool holder in two positions perpendicular to each other on adjacent sided as specified, (see 6.2).

3.8.5 Thread chasing dial. A thread chasing dial shall be furnished which will allow disengaging of the half-nuts and re-engaging during threading. The thread chasing dial shall be mounted on the carriage apron and be driven by the leadscrew. When a variable power feed machine is required a thread chasing dial shall not be required.

3.8.6 Thread chasing stop. The cross slide shall be equipped with a thread chasing stop to ensure repositioning of the tool to the last depth of cut, in preparation for feeding to the next depth of cut.

3.8.7 Drive plate. The drive plate shall mount on the spindle nose and shall have at least one radial slot for driving a lathe dog. Unless otherwise specified (see 6.2) the drive plate shall be the maximum allowable diameter, not to exceed the nominal swing over the bed.

3.8.8 Drill chuck. The drill chuck shall have no less than one-half inch drill capacity and shall mount in the tailstock spindle of the lathe.

3.9 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2). The optional equipment provided shall be functional without requiring modification of the equipment or machine.

3.10 Repair parts. Required repair parts shall be furnished as specified (see 6.2). All repair parts shall be assigned a part number identified in a repair parts manual.

3.11 Marking on instruments, control panels, charts, and plates. All words on instruments, control panels, charts, and plates shall be in the English

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language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant.

3.11.1 Lubrication plate. A lubrication plate shall be permanently and securely attached to each machine. Unless otherwise specified (see 6.2), the plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity Federal or military specification for each lubricant
Size and type of each lubricant filter (where applicable)

3.11.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model number of the basic standard machine and a suffix identified in the manufacturer's permanent records.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Short circuit/over-current protection rating
Contract number or order number
National Stock Number or Plant Equipment Code
Date of manufacture

3.12 Workmanship. Workmanship of the lathe and its accessories shall meet all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall

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become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submissions of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.7.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.6 and the tests in 4.7 which apply to the size lathe ordered. The machine shall pass the examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each lathe prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.6, the tests in 4.7.1 through 4.7.4.3 as applicable to the size lathe ordered. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Examination. Each lathe shall be examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 41^{0} to 104^{0} Fahrenheit and 20 to 95 percent relative humidity.

4.7.2 Operational test. Each machine shall be operated at no load for not less than 1 hour. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.3 Alignment tolerance test. Each machine shall be tested to determine compliance with the alignment tolerance requirements of 3.6. The alignment

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tolerance test shall be performed prior to and subsequent to the performance tests given in 4.7.4. Any machine adjustments required to comply with 3.6 during or after the performance test shall be cause for rejection.

4.7.4 Performance tests. Each lathe shall be subjected to the following cutting tests to determine whether the performance under cutting loads is within the requirements of this specification. All performance tests shall be completed without making any adjustments to or performing any maintenance on the machine.

4.7.4.1 Size 1 and size 2 lathe tests.

4.7.4.1.1 Rough turning test, between centers. A test part of 60,000 pounds per square inch (psi) minimum tensile strength steel, not less than 2 inches in diameter and of sufficient length for chucking and turning a length not less than 12 inches, shall be used. The bar shall be chucked at one end and supported by the tailstock center at the other end. The bar shall be rough turned over a length of not less than 12 inches. The spindle speed, feed rate, and the depth of cut shall be based on the recommended cutting speed for the material and carbide tooling to produce a stock removal rate of not less than one cubic inch of metal per minute per rated horsepower of the main drive motor. There shall be no evidence of stalling or chatter during the cut. Upon completion of the cut, the turned diameter shall be checked for out of roundness and shall not exceed 0.001 inch; and for taper, not to exceed 0.002 inch in 12 inches.

4.7.4.1.2 Finish turning test, between centers. The bar used in 4.7.4.2.1 may be used and turned to have two collars, at least one inch wide, one at each end of the bar. The final finish cut shall be taken over the two collars in one pass using the spindle speed and feed rate recommended for the type material and carbide tooling being used. The finished diameter shall be checked for out of roundness, not to exceed 0.00015 inch; for taper, not to exceed 0.0003 in 12 inches; and surface finish, not rougher than 32 microinches.

4.7.4.1.3 Finish test, part in chuck. A test part of aluminum bar not less than 2 1/4 inches in diameter and of sufficient length for chucking and turning a 2 inch length shall be used. A truing cut may be taken as required. A finish cut at lease 2 inches long shall be made using the maximum practical spindle speed and feed rate recommended for finish turning aluminum bar using carbide tooling. The turned diameter shall be checked for out of roundness, not to exceed 0.000030 inch. Surface finish shall be not rougher than 16 microinches.

4.7.4.2 Size 3 lathe tests.

4.7.4.2.1 Rough turning test, between centers. A test part of 60,000 psi minimum tensile strength steel, not less than 2 inches in diameter and of sufficient length for chucking and turning a length not less than 12 inches, shall be used. The bar shall be chucked at one end and supported by the tailstock center. The bar shall be rough turned over a length of not less than

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12 inches. The spindle speed, feed rate, and depth of cut shall be based on the recommended cutting speed for the material and carbide tooling to produce a stock removal rate of not less than one cubic inch of metal per minute per rated horsepower of the main drive motor. There shall be no evidence of stalling or chatter during the cut. Upon completion of the cut, the turned diameter shall be checked for out of roundness, not to exceed 0.001 inch; and for taper, not to exceed 0.002 inch in 12 inches.

4.7.4.2.2 Finish turning test. The bar used in 4.7.4.3.1 may be used and turned to have two collars, at least one inch wide, one at each end of the bar, a truing cut may be taken as required. The final finish cut shall be taken over the two collars in one pass using the spindle speed and feed rate recommended for the type material and carbide tooling. The finished diameter shall be checked for out of roundness, not to exceed 0.00025 inch; for taper, not to exceed 0.00035 in 12 inches; and surface finish, not rougher than 32 microinches.

4.7.4.2.3 Finish test, part in chuck. A test part of aluminum bar not less than 2 1/4 inches in diameter and of sufficient length for chucking and turning a 2 inch length shall be used. A truing cut may be taken as required. A finish cut at least 2 inches long shall be made using the maximum practical spindle speed and feed rate recommended for finish turning aluminum bar using carbide tooling. The turned diameter shall be checked for out of roundness not to exceed 0.000080 inch. A facing cut shall be made and checked for axial runout, not to exceed 0.000080 inch. Surface finish shall be not rougher than 16 microinches.

4.7.4.3 Additional performance tests. Performance tests required in addition to or in lieu of the requirements given above shall be as specified and fully described (see 6.2).

4.7.5 Electromagnetic interference control test. When specified in paragraph 3.3.8, equipment requiring electromagnetic interference control shall be tested for compliance using the procedures given in MIL-STD-462.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-18058.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The toolroom lathes covered by this specification are intended for use in laboratories, toolrooms, and production shops where the nature of the work requires extreme accuracy, fine surface finish, and high speeds for turning, facing, boring, and threading operations on a variety of materials using modern cutting tools such as carbine and ceramics.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class, and size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. System of units for measuring and indicating devices (see 3.2.1).

f. Digital readout system, if not required (see 3.2.1.2).

g. Specify any special digital readout system features, if required (see 3.2.1.2).

h. Additional safety and health requirements, if required (see 3.2.5).

i. Specify configuration of additional point-of-operation guarding, if required (see 3.2.5).

j. Painting, if different (see 3.3.5).

k. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

l. Headstock temperature sensor, if required (see 3.4.3).

m. Spindle nose, if different (see 3.4.4).

n. Infinitely variable power feed, if required (see 3.4.8).

o. Specify type gear box, if combination English-metric is not required (see 3.4.9).

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p. Tailstock set-over distance, if different; specify amount required (see 3.4.11).

q. Electrical system, if different (see 3.4.12).

r. Motors, if different (see 3.4.13).

s. Worklight, if not required (see 3.4.14).

t. Hour meter, if required (see 3.4.15).

u. Size and capacity, if different (see 3.5).

v. Specify physical size limitation, if required (see 3.5.1).

w. Additional performance features, if required (see 3.7.1).

x. Standard equipment, if different (see 3.8).

y. Specify chuck and chuck jaws required (see 3.8.2).

z. Specify metric graduations if required on the taper attachment (see 3.8.3).

aa. Tool post, specify which type is required (see 3.8.4).

bb. Thread chasing dial, if not required (see 3.8.5).

cc. Specify and fully describe any optional equipment required (see 3.9).

dd. Specify applicable repair parts, if required (see 3.10).

ee. Lubrication plate, if required (see 3.11.1).

ff. Information to be provided on lubrication plate, if different (see 3.11.1).

gg. Nameplate, if different (see 3.11.2).

hh. Inspection conditions, if different (see 4.3).

ii. First article inspection, if different (see 4.4).

jj. Quality conformance inspection, if different (see 4.5).

kk. Additional performance tests, if required (see 4.7.4.3).

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6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
.	.	.	.
.	.	.	.

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.3.1 Technical data. It should be specified in the "Remarks" section of the DD Form 1423 that all technical data ordered shall be written in the English language.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International D'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

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6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

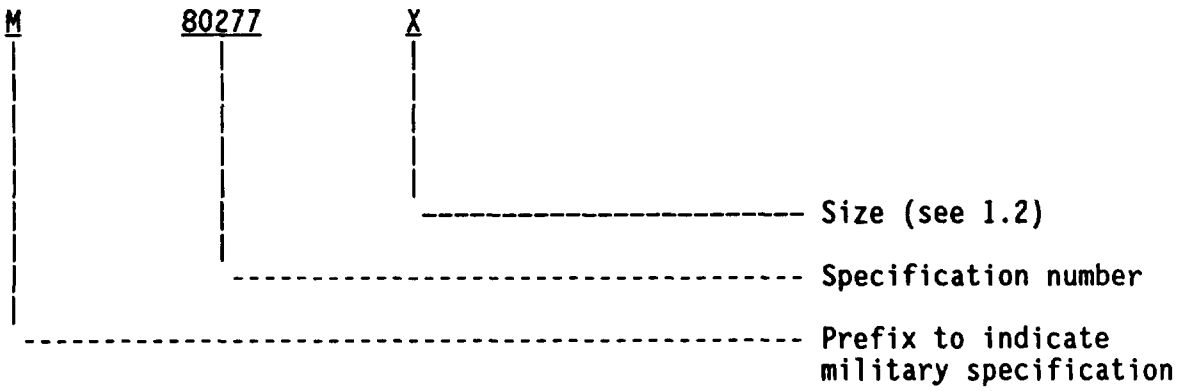
6.8 Safety and health requirements. OSHA 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants, and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

6.9 Training. Training required by the procuring activity should be provided as specified in the contract.

6.10 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.11 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.12 Part or Identifying Number (PIN). The PIN to be used for machines acquired to this specification are created as follows:



6.14 Subject term (keyword) listing.

Turning
Military specification
Precision
Variable speed

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TABLE I. Sizes and capacities.

Characteristics	Size 1	Size 2	Size 3
Capacities:			
Swing over bed	11"	12"	13"
Swing over cross slide	6"	7"	8"
Distance between centers (tailstock fully supported)	18"	20"	25"
Main drive motor	1 HP	5 HP	3 HP
Headstock:			
Spindle type and size	3"-D1 or Mfgr Std	3" - D1	4" - D1
Hole through spindle	1-1/4"	1-3/8"	1-1/2"
Center size, Morse Taper	#2 or Mfgrs Std	#2	#3
Draw-bar collet capacity, round stock	1-1/16"	1"	1"
Optional spindle nose collet, chuck	-----	1-3/8"	1-3/8"
Spindle speed range, variable, (RPM)	125 to 3,000	8 to 3,000	35 to 3,000
Carriage:			
Cross slide travel	5-1/2"	5-1/2"	6"
Compound slide travel	3"	2"	3"
Gear box, English			
Number of feeds	16	50	16
Feed range, inches per revolution (IPR)	0.0005 to 0.20	0.0005 to 0.016	0.0005 to 0.020
(Optional) inches per minute (IPM)	1/4 to 7		
Number of threads	27	60	35
Thread range	11 to 108	3 to 184	4 to 112
Leadscrew diameter and threads per inch (TPI)	1",8	1",8	1",4
Gear box, Combination English and Metric:			
Feed range English	0.0005 to 0.20 IPR	0.0005 to 0.016 IPR	0.0005 to 0.020 IPR
(Optional)	5/16 to IPM		
Metric	0.015 to 0.4mm per rev	0.013 to 0.4mm per rev	0.015 to 0.5mm per rev
(Optional)	8 to 176mm/pm		
Number of threads,			
English,	36	60	35
Metric,	36	60	31
Range of threads, English/	11 to 108 TPI	3 to 184 TPI	4 to 112 TPI
Metric,mm lead	75 to 2.7	0.188 to 11.5	0.1 to 7
Tailstock:			
Spindle diameter	1-3/8"	1-3/4"	1-5/8"
Spindle travel	3-1/2"	3-1/2"	4"
Center size, Morse Taper	#2	#2	#2

NOTE: Machine size and capacities shall be not less than the stated requirements of the size ordered nor greater than the requirements of the next larger size shown in above table. When the largest size and capacity machine is ordered the size and capacity of the machine offered shall not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the minimum or less to maximum or greater.

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TABLE II. Alignment and performance accuracies.

ANSI B5.16 test number	Test (Dimensions in inches)	Toolroom lathe
1	Bed level, transverse direction, in 12 inches	±0.00024
2	Bed level, longitudinal direction, in 12 inches	within 0.00024
3	Tailstock way alignment, in 12 inches	within 0.00020
4	Spindle center runout	0 to 0.00025 TIR
5	Spindle nose runout	0 to 0.00025 TIR
6	Cam action of spindle	0 to 0.0003 TIR
7	Spindle taper runout, at end of 12-inch test bar	0 to 0.0006 TIR
	at end of spindle nose	0 to 0.0003 TIR
8	Headstock alignment vertical, in 12 inches	0 to 0.0004
9	Headstock alignment horizonal, in 12 inches	0 to ±0.0002
10	Tailstock spindle alignment horizontal in 4 inches	0 to 0.0004
11	Tailstock spindle alignment vertical, in 4 inches	0 to 0.0005
12	Tailstock taper alignment horizontal, in 12 inches	0 to ±0.0004
13	Tailstock taper alignment vertical, in 12 inches	0 to 0.0004
14	Vertical alignment of head and tail centers, high at tailstock	0 to 0.001
15	Readings for base length bed taken with lead screw stationary: Vertical	0 to 0.003
	Horizontal	0 to 0.003
16	Alignment of half-nut horizontal or vertical	0 to 0.006
17	Cross-slide alignment, on 12-inch diameter	0 to 0.0002
18	Face plate runout On diameter	0 to 0.0005
	On face at diameter	0 to 0.0005
19	Chuck runout Face and periphery	0 to 0.003
	Face of steps	0 to 0.001
	Bar test 3 inches from end of jaw	0 to 0.003 maximum
20	Collet chuck runout 1 inch from collet chuck	0 to 0.001
21	Lathe must turn round with work mounted in chuck	0 to 0.0002
22	Lathe must turn cyclindrical, 12-inch workpiece mounted in a chuck	0 to 0.0006
23	Lathe must turn cylindrical, 12-inch workpiece mounted on centers	0.0004
24	Leadscrew, lead per 12 inches	±0.001
	lead in any 4 inches	±0.0004
25	Backlash on cross-feeds screw	0.003
	on compound rest screw	0.003

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Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Air Force - 84

User activities:
Navy - MC

Preparing activity:
DLA - IP

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